EEII I
Drive Electronics

Work Package Scope of Work — Engineering Design 2026

CONSTRAINT: Protection against shoot-through and overcurrent is mandatory

Main Work Elements

1. **Full VFD chain design —** a simple rectifier and bulk capacitors for the HV DC bus, plus single-phase-to-3-phase inversion (discrete MOSFET/IGBT with gate drivers) driving the induction motor. Architecture is largely carried over from 2025's approach.
2. **Component selection —** switching devices, gate driver ICs, and a current-sense resistor feeding the shared ADC channel.
3. **Protection circuit design and validation —** the Increment 2 half-bridge PCB milling exercise — students mill a board from a supplied component set and use it to validate shoot-through and overcurrent protection before committing to the full 3-phase board.
4. **Simulation —** switching behaviour, protection response, and thermal headroom ahead of hardware commitment.
5. **E-stop integration —** a physical e-stop wired directly into the HV bus so activation reliably kills the system's main dangerous component.
6. **System power integration —** the main on/off switch and mains distribution to the HV bus and to each subsystem's already-selected COTS low-voltage supply. Integration and switching only — each work package selects its own supply.
7. **Controller interface —** the six isolated gate/PWM lines plus reserve line; current-sense conditioned to a single-ended signal capped below 2.048V (the shared ADC's internal reference), occupying one ADC channel; fault inferred in firmware from commanded-vs-measured mismatch rather than a dedicated hardware flag.
8. **Full build and test —** assembling the complete VFD board, verifying it drives the induction motor, confirming protection trips under fault injection, and verifying e-stop and power distribution.

Scoped for effort parity with the other four 2026 work packages (MCTR I, MCTR II, EEII I, EEII II, REII).